



DANIEL A FARONBI

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SUMMARY

I'm a Music Technology Researcher, Jazz Pianist, and Music Producer. My research involves using signal processing and machine learning techniques for the generation and analysis of music and other audio signals. This includes research in areas like Generative Audio Models, Music Information Retrieval, Computational Musicology, and more.

WEBSITES, PORTFOLIOS, PROFILES

- <https://danielfaronbi.com/>
- <https://www.linkedin.com/in/daniel-faronbi-7aa61b184/>
- <https://github.com/dafaronbi>

SKILLS

- Python, C++, C, Matlab, Java, Java Script, Kotlin
- Machine Learning/Deep Learning
- Generative Audio Models
- Computational Musicology
- Digital Signal Processing,
- Computer Vision, Music Information Retrieval
- Embedded system development: ARM, AVR, and PIC microcontrollers.
- Linux and Minix Kernel Knowledge
- Front and Back end web development (java script, node js, django)
- PCB design with Eagle Cad and OrCad
- Serial communication protocols: SPI, I2C, UART
- Electronic Circuit Design: amplifiers, voltage regulators, current sources
- Electrical skills with Logic Analyzer usage, Multisim, Oscilloscope usage, Soldering
- Production in Ableton Live, FL Studio, Max MSP, and Pro Tools

EXPERIENCE

ATG Intern / Dolby Laboratories - San Fransico, CA

06/2023 - 09/2023

Audio research as part of the Advanced Technology Group (ATG). Worked on applications for streaming Dolby Atmos audio for collaboratively mixing audio.

Machine Learning Engineering Intern / Bose Corporation - Farmingham, MA

06/2022 - 08/2022

Worked on projects including environmental sound based compositions, source separation for speech, music, and sound effects, and evaluation techniques for deep RMS noise estimators.

Research Assistant (Operations Research) / University Of Nebraska At Omaha - Omaha, NE

08/2020 - 05/2021

Research Assistantship of NSF funded project through the mathematics department at UNO. The research is focused on implementation of multidimensional search algorithms for linear programming with applications in the fields of machine learning or data science.

Wearable Computing REU / Washington State University - Pullman, WA

05/2020 - 08/2020

One NSF's prestigious Research Experiences for Undergraduates (REU) programs. The research focused on using DSP

and machine learning techniques to classify stressed and unstressed regions in a variety of biometric signal data. The classifications of these signals were used to indicate the likelihood of alcohol relapse.

Engineer Intern / ScanMed LLC - Omaha, NE

07/2019 - 08/2020

Computer vision and machine learning work centered at creating automated diagnosis for prostate. Embedded microcontroller development for controlling MRI coils.

Media Support Engineer / Boys Town National Research Hospital - Omaha, NE

02/2018 - 12/2018

Provided audio and visual support for a variety of experiments at Boys Town National Research Hospital.

Network Engineer Intern / Union Pacific - Omaha, NE

06/2017 - 12/2017

Provided server connectivity for Union Pacific's network applications using CISCO IOS to configure switches and routers and editing perl scripts used to generate DNS configurations.

TEACHING EXPERIENCE

Music Technology Adjunct Faculty

(MPATE-GE 2617) C Programming for Music Technology Lab:

- This course provides students with hands-on instruction in algorithmic design & general programming concepts. Example code is implemented, discussed, & demonstrated in detail in order to provide students with step-by-step examples of programming design paradigms

(MPATE-GE 2639) Adv Topics Music Tech: C++Audio Application Development Lab:

- This is an advanced graduate-level course covering the C++ programming language, with an emphasis on developing applications for music and audio. Students are expected to have working knowledge of C; this course extends that knowledge to C++, exploring object-oriented topics such as classes, inheritance, function and operator overloading, polymorphism and encapsulation. Students use existing frameworks and libraries to create their own audio applications and audio plug-ins.

Sound + Science Student Mentor

Sound+Science is a Simons Foundation funded innovative after-school program, research lab, and community in NYC for high school students interested in exploring the unique and deep connection between music, science, and technology. Through a unique curriculum, participants will explore modules like 'The Hidden Codes in Electronic Music', 'From Hip-Hop to AI', and Music Improvisation and the Evolution of the Mathematics of Computation Sound+Science, revealing the profound connections between music, technology, physics and mathematics. <https://soundplusscience.com/>

EDUCATION AND TRAINING

Ph.D.: Music Technology

New York University - New York, NY

GPA: 3.422

Bachelor of Science: Computer Engineering

05/2021

University of Nebraska At Lincoln

GPA: 3.658

Bachelor of Arts: Music Technology

05/2021

University of Nebraska At Omaha

Minor: Mathematics

GPA: 3.658

ACTIVITIES AND HONORS

UNO Walter Scott Jr. Scholarship Recipient

The Walter Scott, Jr. Scholarship Program (Scott Scholars) is the most prestigious scholarship extended by the University of Nebraska system. The scholarship provides complete funding for an undergraduate college experience that includes tuition, room and board, and whole-person learning opportunities. Eligible areas of study include majors within the UNL College of Engineering, UNO College of Information Science and Technology, and other STEM related programs.

UNO Regent's Scholarship Recipient

Full tuition scholarship

UNO Honors Program Member

The University Honors Program is an academic learning community. The mission of the Honors Program is to provide an enhanced and supportive learning environment responsive to the educational needs of highly able and/or exceptionally motivated undergraduate students.

IEEE Eta Kappa Nu - IEEE National Honor Society

IEEE-Eta Kappa Nu (IEEE-HKN), the honor society of IEEE, promotes excellence in the profession and in education with ideals of Scholarship, Character and Attitude. It was founded on 28 October 1904 for students of electrical engineering. Today, IEEE-HKN promotes professional accomplishment, service, and development for students and professionals in electrical engineering, computer engineering, and other IEEE fields of interest.

EXTRACURRICULAR EXPERIENCE

NYU GenAudio & AI Club

The NYU GenAudio & AI club is a group of students and professors who are interested in a variety of applications of generative AI music models. Members of the group read cutting edge papers on generative AI, hear talks from industry and academic experts on AI, and work on relevant group projects focused on a variety of subtasks.

National Association of LSAMP Alumni (NALA)

NALA's Vision is to ensure that underrepresented minority students in STEM are prepared to assume leadership roles and professional opportunities. We will foster the career development and transition of our LSAMP students through conferences, workshops, and mentorship. We will do this by developing a network of LSAMP Alumni to shine a brighter future for our current undergraduate students.

Walter Scott Junior Leadership Council

This was a leadership position for high achieving STEM students admitted to the Walter Scott Jr. Scholarship program. We helped revamp the mentorship aspect of the program to include upperclassmen mentoring lowerclassmen. We redesigned the admission process to be more holistic rather than focus on only set academic achievements. Harvard design thinking methods were used to solve problems.

Prairie STEM Teaching Assistant

Daniel has volunteered for the after school elementary STEM education program, Prairie STEM. In this program, young students are exposed to a variety of engineering projects. They had experience using CAD software to design architecture, mechanics, and circuits. Daniel volunteered as a teaching assistant for these lessons.

Music

Daniel is active jazz pianist, music producer, and composer. He has performed for music groups spanning a wide variety of genres and has received both jazz and classical training. He has worked as a studio manager for Studio 381 at the University of Nebraska and has published two albums. He has shared the stage with jazz heavy hitters such as Benny Golson, Bobby Sanibria, Micah Bell, and Dave Stryker and performed at world famous jazz venues such as The Django and Nublu. He maintains an active performance schedule at various venues in New York City.

RELEVANT COURSES

- 3D Audio
- Audio Recording Techniques I, II, & III | Advanced Computer Music Composition
- Calculus I, II & III
- Communication Systems | Computer Vision
- Data Structures | Deep Learning | Differential Equations

- Digital Design | Digital Synthesis
- Electric Circuits I & III | Electronic Circuits I and II
- Embedded Microcontrollers | Finite Difference Equations
- Fundamentals of Conducting |
- Information Theory | Introduction to Data Science
- Java I and II | Jazz Arranging and Composition
- Psychology of Music
- Linear Algebra | Machine Learning
- Microprocessor Applications
- Microprocessor System Design | Mobile Robotics | Music History I and II
- Music Informatics | Music Information Retrieval
- Music Theory (3 classes) | Operating Systems
- Physics I and II (with calculus)
- Sound Reinforcement | Statistics
- Switching Circuits

ACADEMIC SERVICES

Conference Reviewing

- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
- International Joint Conference on Neural Networks (IJCNN 2025)

PUBLICATIONS

- **Faronbi, Daniel**, Iran Roman, and Juan Pablo Bello. "Exploring Approaches to Multi-Task Automatic Synthesizer Programming." *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.
- **Faronbi, Daniel**; Brown, Phil; 2024; Remote Collaborative Mixing with 3D Audio [PDF]; Music and Audio Research Lab, New York University; Dolby Laboratories; Paper 10708; Available from: <https://aes2.org/publications/elibrary-page/?id=22521>
- Iran Roman, **Daniel Faronbi**, Isabelle Burger-Weiser, Leila Adu-Gilmore. "F0 ANALYSIS OF GHANAIAN POP SINGING REVEALS PROGRESSIVE ALIGNMENT WITH EQUAL TEMPERAMENT OVER THE PAST THREE DECADES: A CASE STUDY" *SMC 2023 - Sound and Music Computing Conference 2023 (SMC)*.
- McFee, B., Matt McVicar, **Daniel Faronbi**, Iran Roman, Matan Gover, Stefan Balke, Scott Seyfarth, Ayoub Malek, Colin Raffel, Vincent Lostanlen, Benjamin van Niekirk, Dana Lee, Frank Cwitkowitz, Frank Zalkow, Oriol Nieto, Dan Ellis, Jack Mason, Kyungyun Lee, Bea Steers, ... Waldir Pimenta. (2024). *librosa/librosa: 0.10.2.post1 (0.10.2.post1)*. Zenodo. <https://doi.org/10.5281/zenodo.11192913>

DATASETS

- **Daniel Faronbi**. (2023). Multi-Task Automatic Synthesizer Programming Preset Dataset [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.7686668>